Six Months with AMD Strix Halo

Local AI, Open Source, and Hardware Reality

A Six Month Review of a Strix Halo Rig

AMD Strix Halo | Local AI | Home Assistant | 3D Modeling | Open Source

winmutt | June 2026
github.com/winmutt

What We're Covering

Section 1: The Hardware Stack

- Power Consumption: Corsair 300 vs RTX 5090 vs Cloud
- The Hardware: AMD Ryzen AI Max+ 395 APU
- The Stack: ROCm + Lemonade Server
- Issue #1070: Core affinity across dual CCDs

Section 2: Projects & Applications

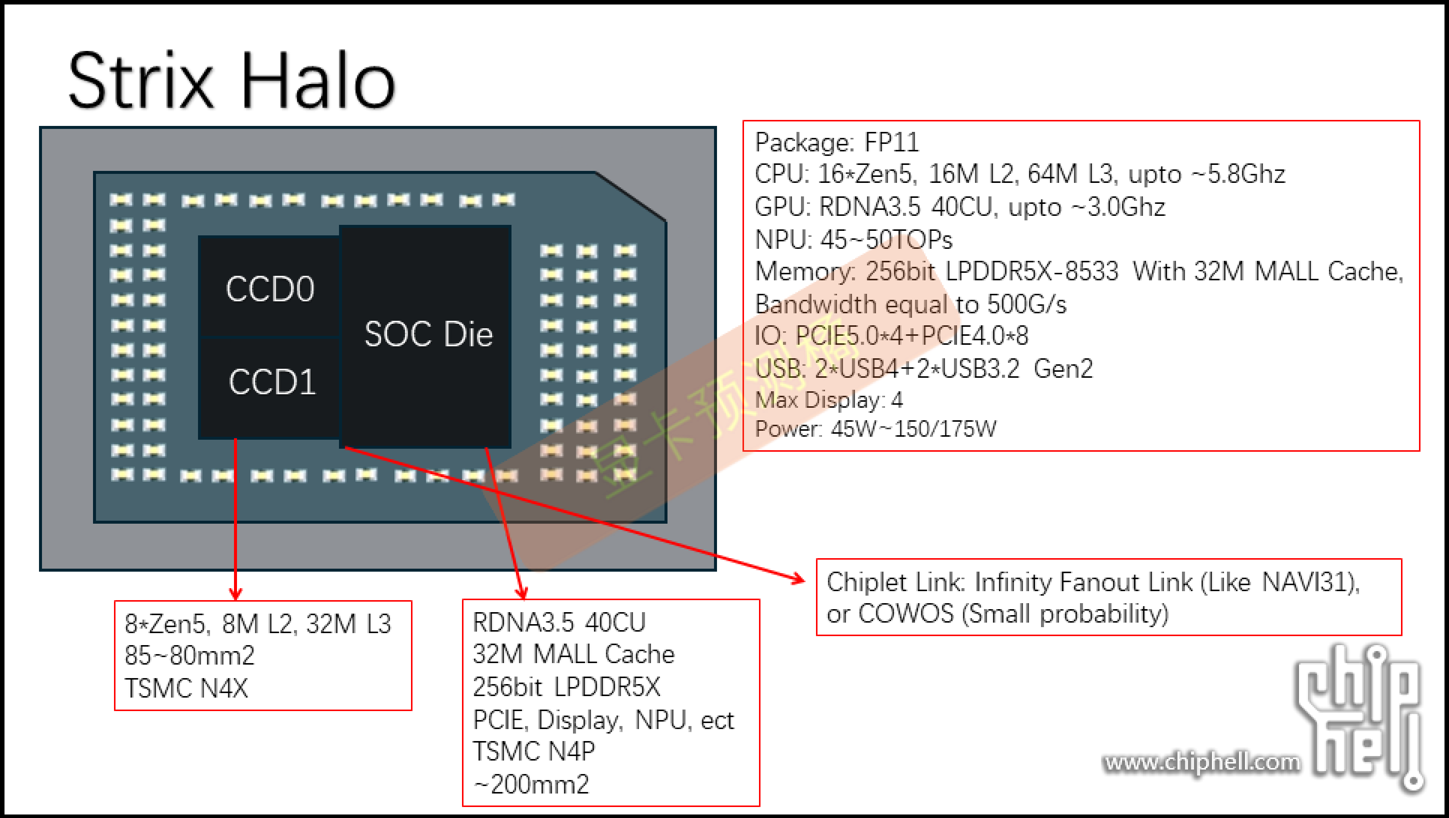
- Home Assistant: Cutting the Alexa cord
- 3D Modeling: AI to concrete signs
- AI Coding Editors: Opencode, Cline, Aider
- Project WWS: Remote workspace provisioning

Section 3: Open Source Journey

- Lessons Learned: What worked, what didn't
- Key Insights: 8 takeaways from 6 months
- OSS Contributions: 3.5x increase
- 26 Years: From LKML 2000 to Strix Halo

The Hardware: AMD Ryzen AI Max+ 395

APU Die Architecture



Physical Layout

- 16 cores (12 performance + 4 efficiency)
- 2x Core Complex Dies (CCD) with 32MB L3 cache each
- 768KB L1d + 512KB L1i + 16MB L2
- Integrated RDNA 3.5 GPU (40 Compute Units)

• 128GB LPDDR5X-8000 unified memory (soldered)
• Source: AMD Ryzen AI Max+ 395 teardown analysis (Chiphell, TechPowerUp)

AMD Radeon 680M GPU

• RDNA 3.5 architecture, 40 Compute Units
• Peak performance: ~4.6 TFLOPS
• Shared memory with CPU (128GB unified)
• Supports ROCm for AI/ML workloads
• Source: AMD Ryzen AI Max+ 395 specifications

AMD XDNA 2 NPU

• Neural Processing Unit for AI workloads
• Integrated into Ryzen AI Max+ 395 APU
• Supports FastFlowLM for efficient inference
• Power-efficient alternative to GPU processing

FastFlowLM Performance (June 2026):

• 6784 tokens prefill in ~15 seconds
• ~455 tokens/sec on NPU
• Chunked processing: 4096 + 2688 tokens
• End-to-end (prefill to response): ~16.2 seconds
• Source: FastFlowLM logs (lemonade-server)

Power Consumption Reality

System Comparison

System	Power Draw	Notes
AMD Strix Halo (Corsair 300)	~300W	Single APU, 128GB unified memory
RTX 4090 Setup	~600-800W	GPU (450W TDP) + CPU + system
RTX 5090 Setup	~800-1000W	GPU (600W TDP) + CPU + system
Commercial AI Server	1500-3000W	Multi-GPU (A100/H100: 400-700W each)

Strix Halo: ~300W vs RTX 5090: ~800-1000W vs Cloud Servers: 1500-3000W
Source: Corsair specs, NVIDIA TDP ratings, TechPowerUp benchmarks

The Hardware Investment

Black Friday 2025 Deal

• Corsair 300 AI Workstation: **$1800** (128GB Strix Halo system)
• Sold NVIDIA RTX 3060 rig to fund the purchase
• Today's equivalent: ~$3000+ (RTX 4090 configurations)
The Verdict: "The Strix Halo was a bargain at $1800 for what it delivers - today you'd pay 2x that for equivalent performance with discrete GPU"

Feat #1070: [enhancement] Core affinity when running multiple models.

Filed
February 8, 2026
Repository
github.com/lemonade-sdk/lemonade/issues/1070
Status
Open

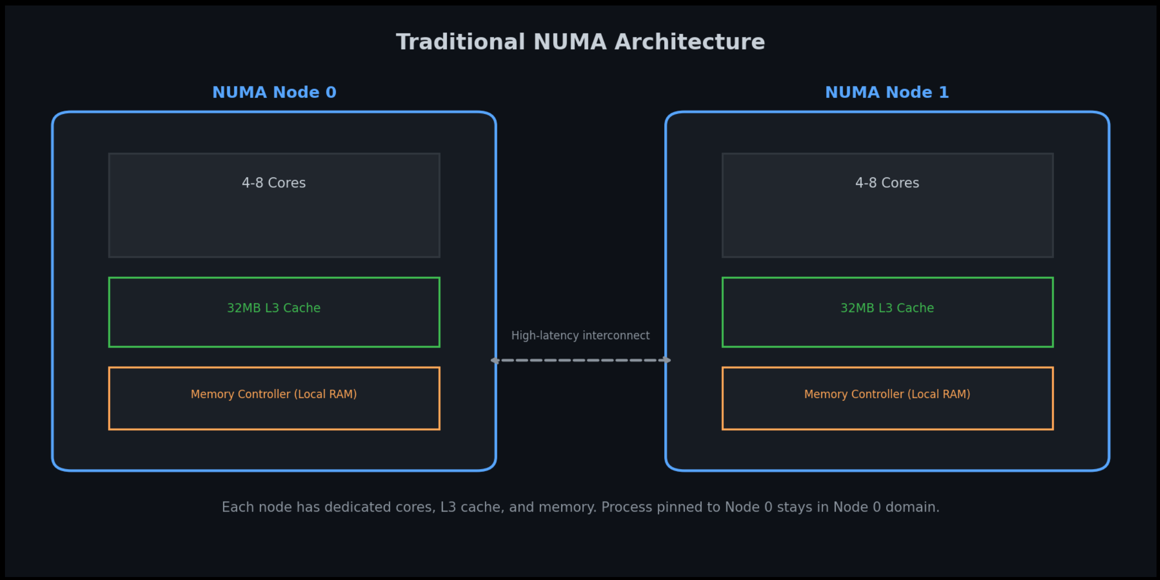
Hardware Reality

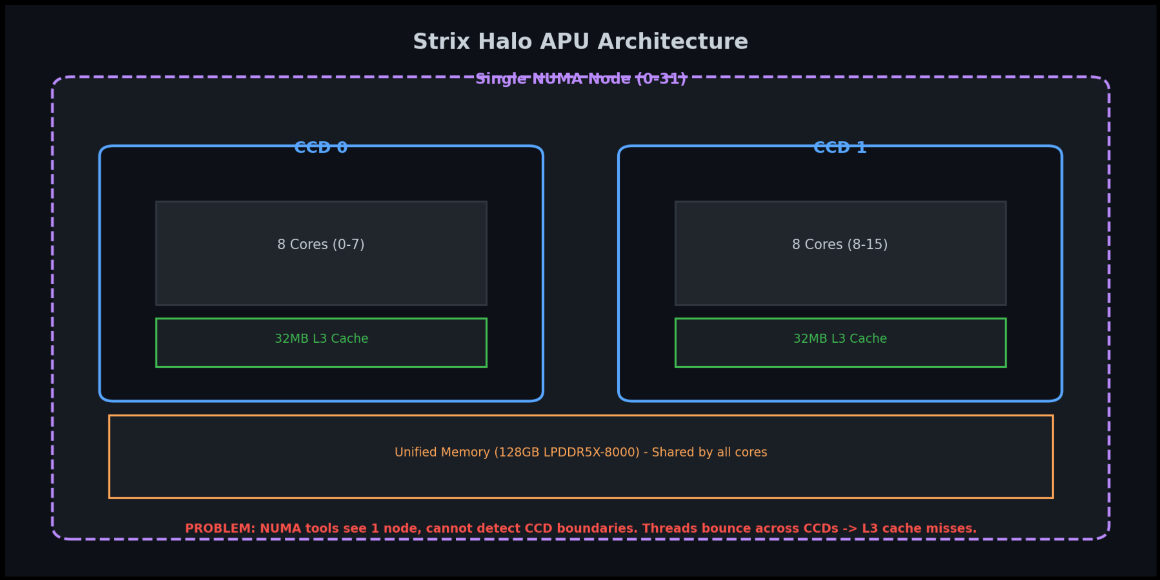
• One processor, but 2x core dies (CCDs) with separate 32MB L3 caches

• Traditional NUMA: 2 nodes, each with own cores + L3 cache

• Strix Halo: Single node, but 2 CCDs with separate L3 caches

Traditional NUMA vs. Strix Halo APU





The Problem: NUMA Tools Can't See CCD Boundaries

- Traditional NUMA tools (numactl, NUMATopologyFilter) detect only 1 node
- Cannot see CCD boundaries automatically
- Manual core pinning required to keep workloads on same CCD
- Running multiple LLMs: threads bounce across CCDs → L3 cache thrashing

Result: Cross-CCD traffic → slower inference, higher latency

The Solution (Proposed)

Manual core pinning (like Red Hat's vcpu_pin_set approach):

- Reserve cores per CCD, pin llama-server accordingly
- Pin based on number of models loaded and CCD boundaries
- Prevent cache-crossing penalties

Result: Each model stays in its own L3 cache domain → no cross-CCD traffic

Evidence: Threads Bouncing Across CCDs

ps Output: Threads Bouncing Across CCDs

PID	LAST_CPU	AVG_MHZ	COMMAND
56840	6	55.7	llama-server (Model 1)
56840	20	54.2	llama-server (Model 1)
59798	0	51.2	llama-server (Model 2)
59798	2	47.5	llama-server (Model 2)
56840	22	55.7	llama-server (Model 1)
56840	4	54.2	llama-server (Model 1)
59798	1[illegible]	51.2	llama-server (Model 2)
59798	18	47.5	llama-server (Model 2)

Notice: LAST_CPU jumps across CCD boundaries

CCD0: cores 0-7, 16-23 | CCD1: cores 8-15, 24-31

Hardware Topology: CCD Boundaries (lscpu)

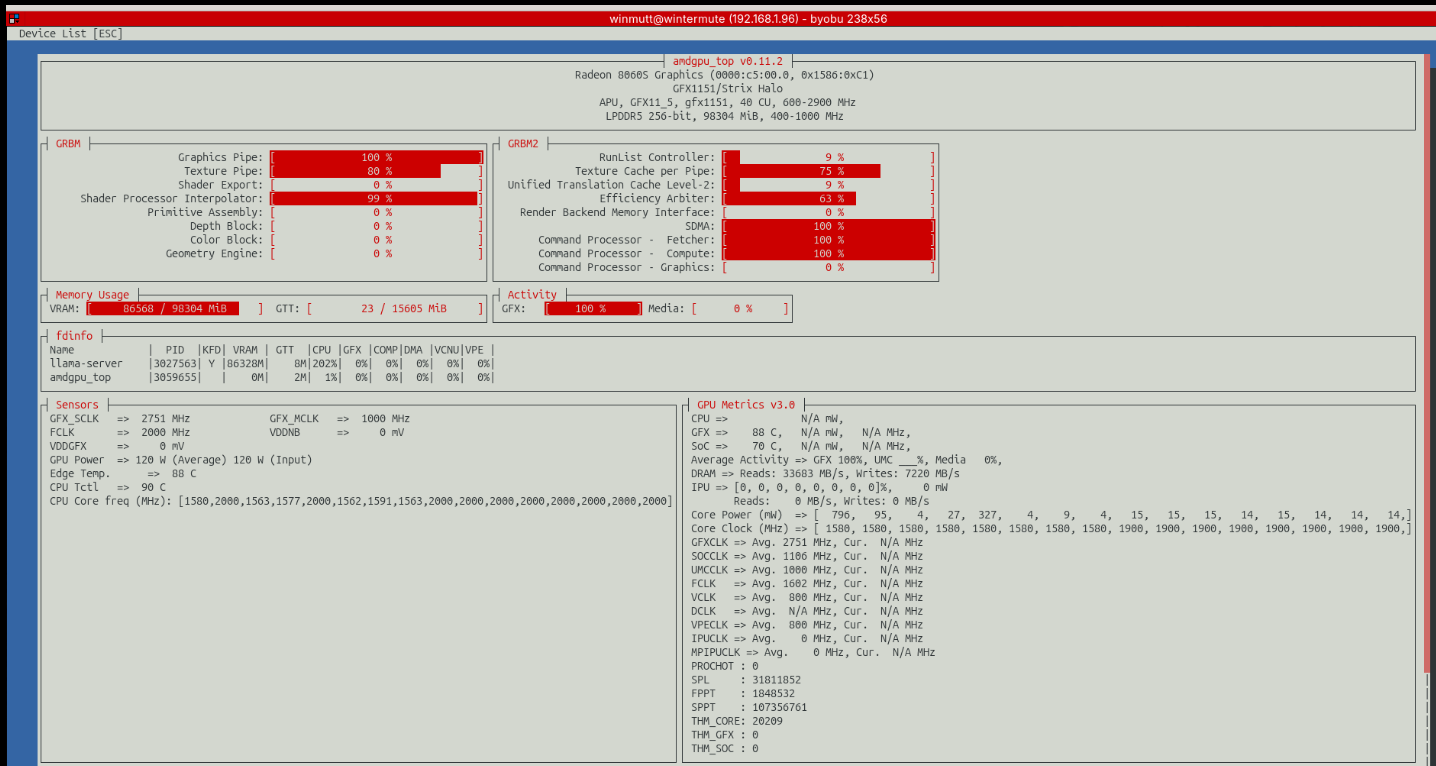
lscpu Output: CCD Topology

CPU	NODE	CORE	L3	MAXMHZ	AVG_MHZ
0	0	0	0	5187.0	4937.8
1	0	1	0	5187.0	2000.0
2	0	2	0	5187.0	4909.0
3	0	3	0	5187.0	4904.9
4	0	4	0	5187.0	4889.9
5	0	5	0	5187.0	4845.2
6	0	6	0	5187.0	4882.0
7	0	7	0	5187.0	4950.4
8	0	8	1	5187.0	2000.0
9	0	9	1	5187.0	2000.0
10	0	10	1	5187.0	2000.0
11	0	11	1	5187.0	2425.1
12	0	12	1	5187.0	3056.8
13	0	13	1	5187.0	4661.1
14	0	14	1	5187.0	2000.0
15	0	15	1	5187.0	2000.0

L3 Cache + AVG_MHZ: Shows load shifting across CCD boundaries

NUMA tools see: 1 node (0-31) | Reality: 2 CCDs with separate L3 caches

Evidence: amdgpu_top



Cutting the Alexa Cord

LineageOS on Echo

December 2025
Started exploring XDA Forums for Echo device hacking

The Process:

1. Unlock bootloader via amonet exploit

github.com/R0rt1z2/amonet

2. Install TWRP recovery

3. Flash LineageOS 18.1 (Android 11)

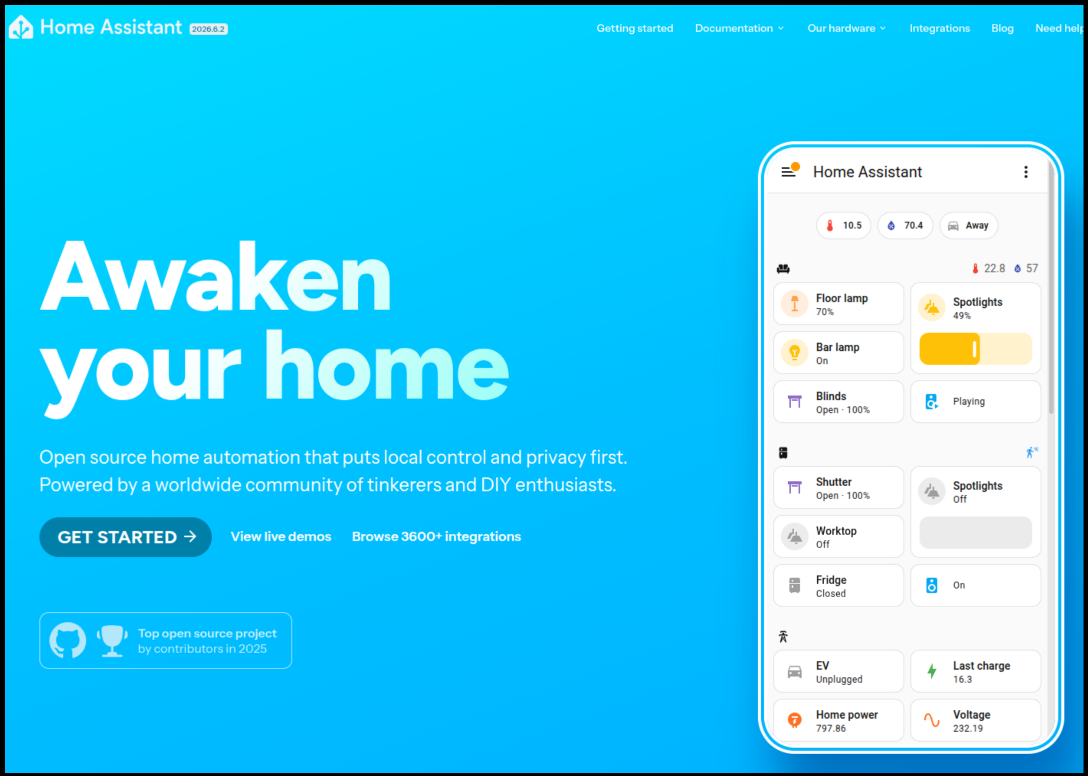
4. Configure ViewAssist for Home Assistant

dinki.github.io/View-Assist

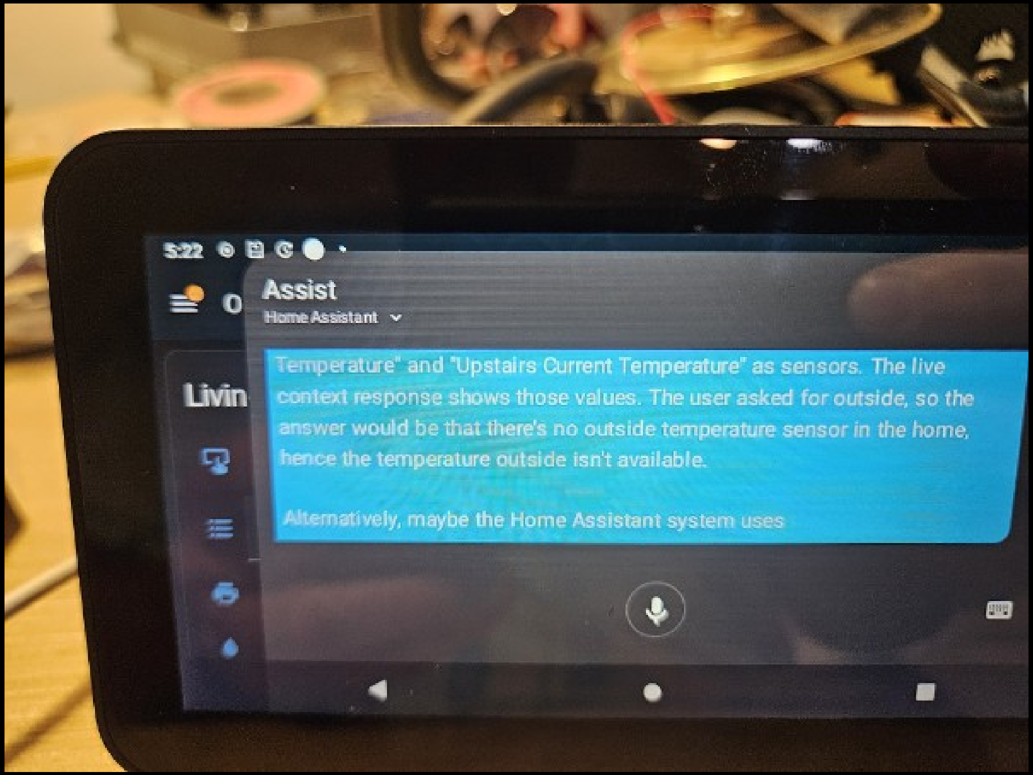
> ***Philosophy***
> *"Basically anything with a USB port. If there is a port, there is a way."*

Home Assistant Dashboard

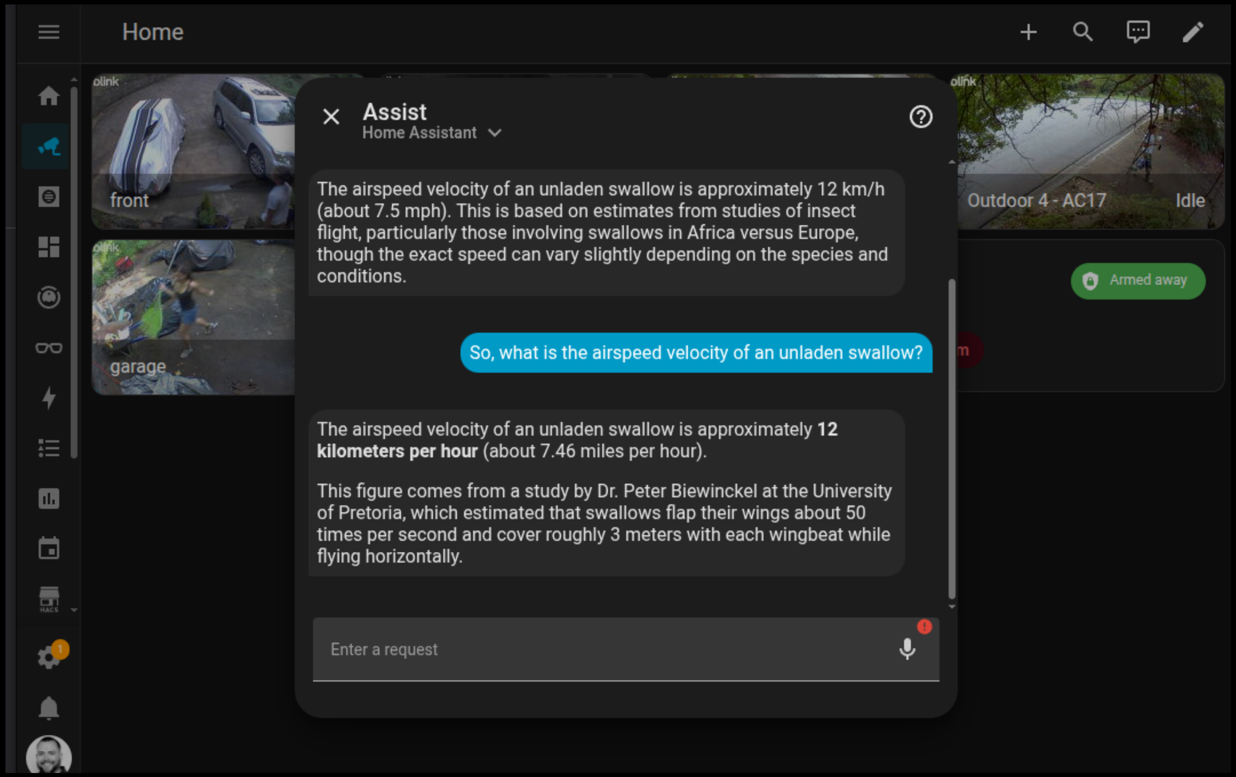
The best open source project I have seen



Home Assistant on Echo Show



Home Assistant Setup



LineageOS Device Compatibility

Echo Show Devices Supported

Device	Codename	SoC	XDA Thread
Echo Show 5 (1st Gen 2019)	checkers	MT8163	xdaforums.com/t/rom-unofficial-11-checkers...
Echo Show 5 (2nd Gen 2021)	cronos	MT8163	xdaforums.com/t/rom-unofficial-11-cronos...
Echo Show 8 (1st Gen 2019)	crown	MT8163	xdaforums.com/t/rom-unofficial-11-crown...

All devices use MediaTek MT8163 SoC

LineageOS 18.1 (Android 11) - community builds by @R0rt1z2

Alexa Replacement Timeline

Dec 6	Started exploring XDA forums
Dec 7	HA running on Echo devices
Jan 19	HA setup on Echo Show Gen 2
Jan 21	Everything works (except wake word)
Jan 25	Hey Jarvis to the rescue
Jan 29	End-to-end complete
Jan 7, 2026	Issue #4: Echo 8 mic dies after 24h
Feb 2026	Still debugging (it's fine)

Issue #4: The Echo 8 Mic That Quit

Audio stops working after ~24 hours

Filed
January 7, 2026
Repository
https://github.com/amazon-oss/releases/issues/4
Status
Open (still waiting, it's fine)

The Problem

• Echo 5: Stable for days

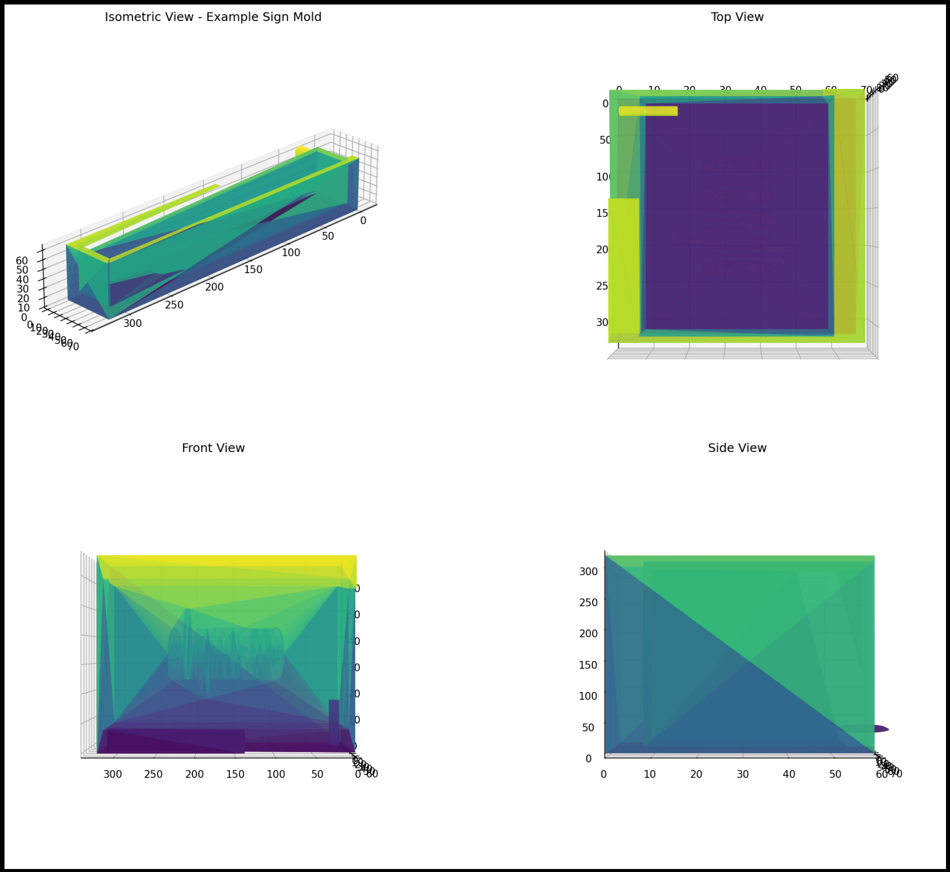
• Echo 8: Dies after ~24 hours

• *Root cause: Unknown (yet)*

Blender: Concrete Sign Mold Design

AI-Assisted Physical Design

Blender + AI workflow for creating physical objects



The Process

1. AI generates Blender design
2. Export to STL
3. 3D print mold
4. Create silicone mold
5. Cast concrete

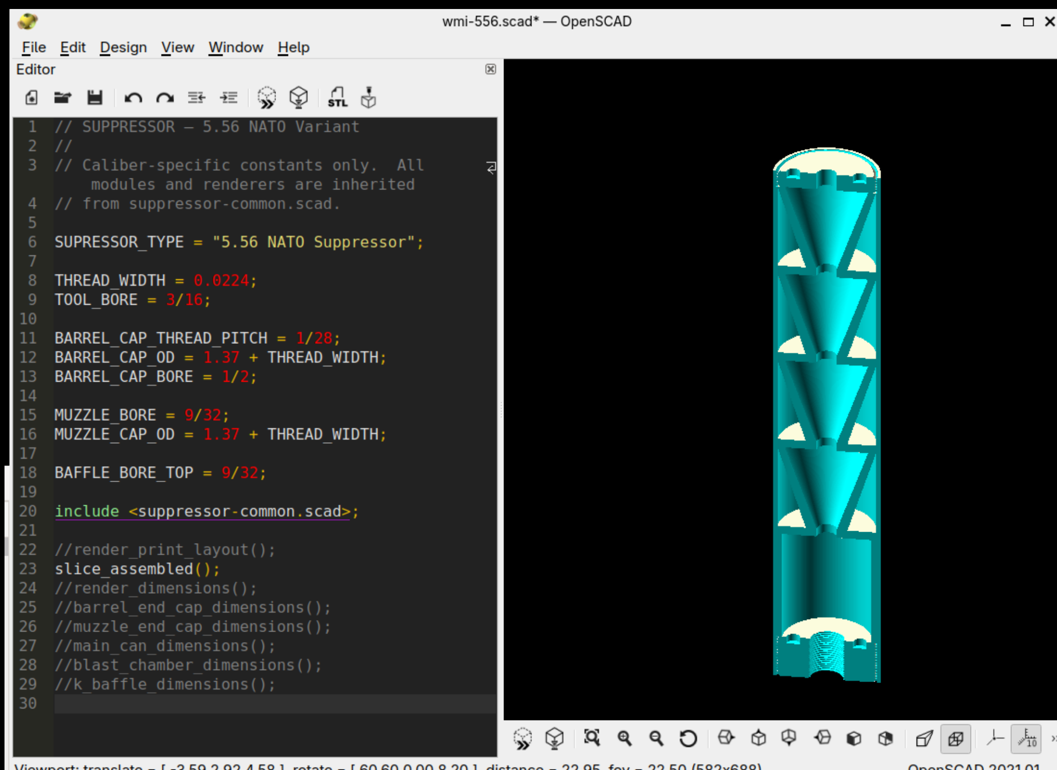
December 1, 2025 — More physical projects coming soon

OpenSCAD: Where Things Work

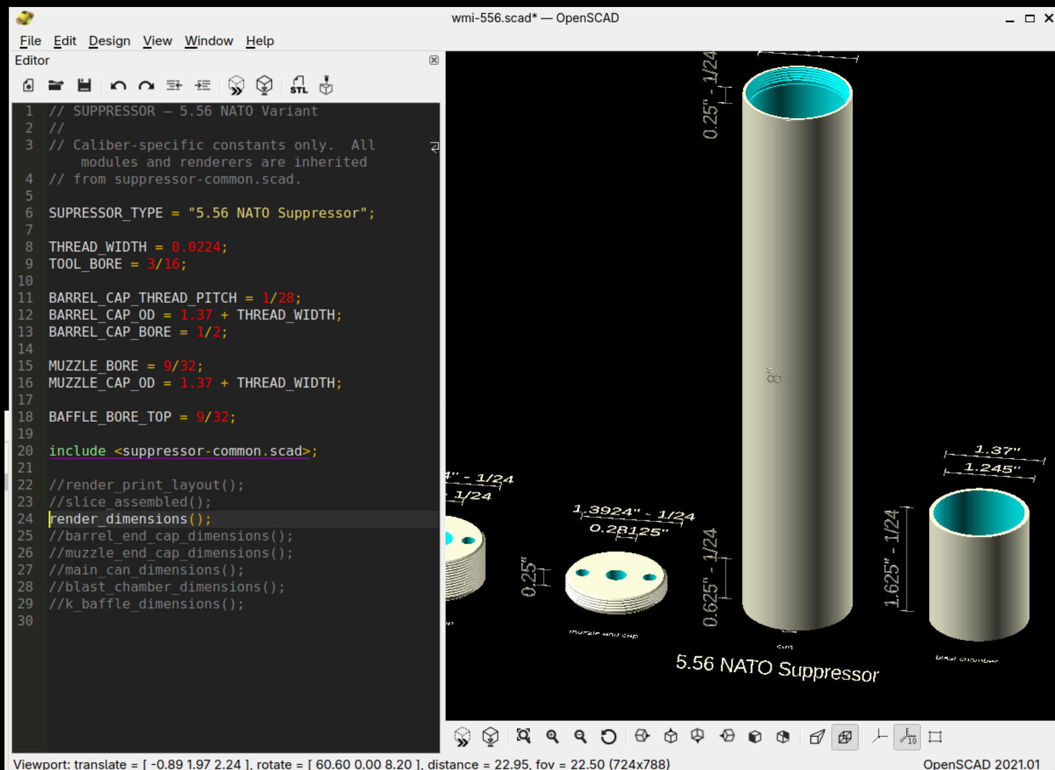
wmi002 Session — Precision Engineering

Success Story
- Parametric design: Caliber-specific constants (.308 Winchester, 5.56 NATO)
- Modular architecture: include , (BOSL2)
- Precision engineering: Thread pitches (5/8-24 UNF), bore dimensions, tolerances
- Working output: K-baffles, blast chambers, threaded end caps

K-Baffle Assembly (Internal View)



Dimensioned Parts View



Two Approaches to Physical Design

Project WWS: I Vibe-Coded an Entire System

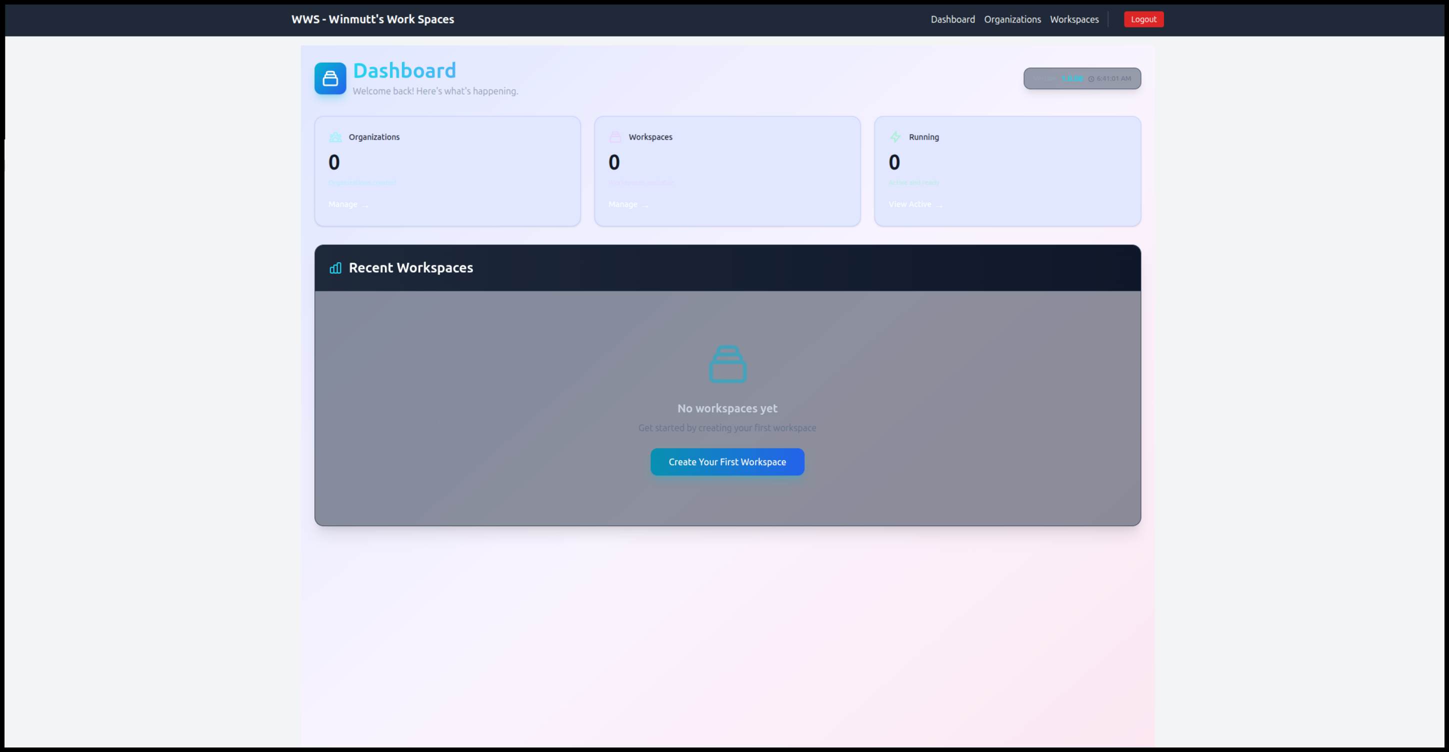
Winmutt Work Spaces — github.com/winmutt/wws

> ***May 12, 2026***
> *"Definitely no Athena and its taken months to get there but this is something I entirely vibe coded."*

Remote workspace provisioning with KVM/Podman isolation

code-server (VSCode in browser) + GitHub OAuth + RBAC

First commit: Feb 22, 2026 → Production: May 12, 2026



AI Coding Editors: My Daily Drivers

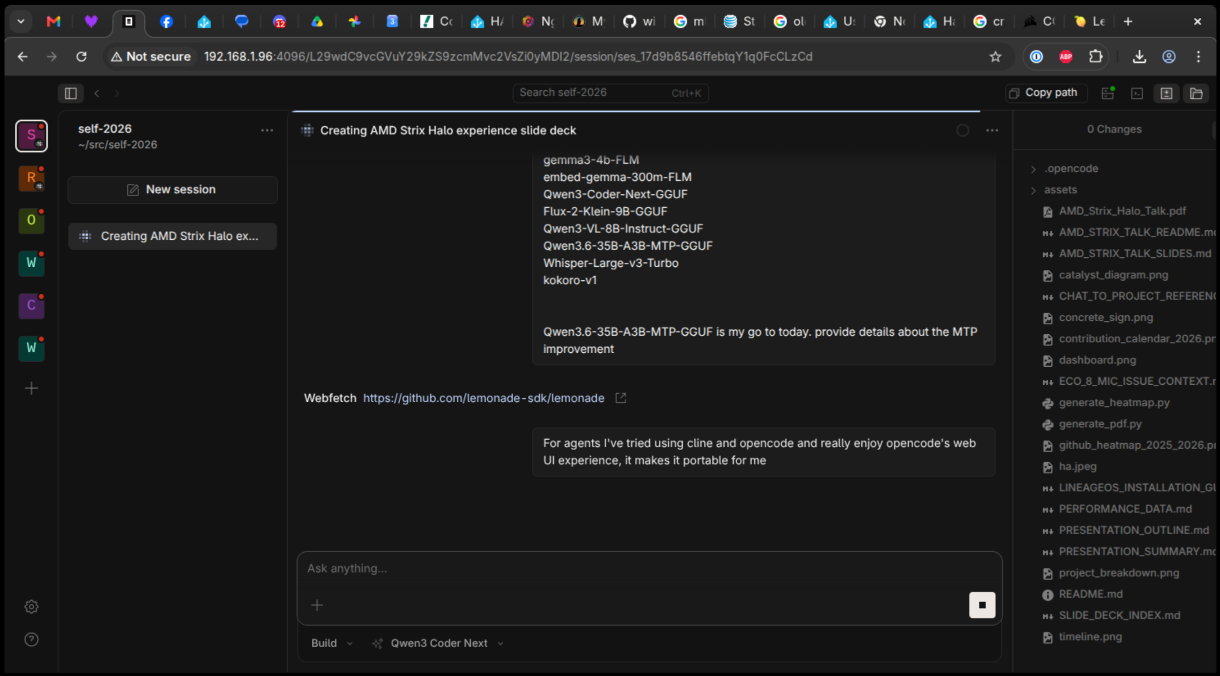
Opencode

Most usable, portable, and stable. Best agentic coding experience. Runs as a web server (http://localhost:3000) allowing remote access with a localized agent running for autonomous development. github.com/anomalyco/opencode

Cline

Good TUI but many regressions over time.

Also Use: Koo Roo, Aider



Lessons Learned (Mostly)

What Worked ✓

• Cline + Qwen3-Coder: Actually produces code
• Prompt Caching: Things hum now
• AMD GPU Libraries: 2x faster (worth the pain)
• WWS Project: Vibe-coded from Feb 22 (first commit) to May 12
• Home Assistant: Alexa is dead, long live OSS

What Didn't ✗

• Memory: 32GB reserved for Lemonade/llama.cpp/opencode
• Ollama: Still crashes after updates
• NPU: Coming soon™ (always)
• Context >64k: TPS drops like a stone

1. Hardware hype ≠ reality (APU challenges are real)
2. Software maturity: bleeding edge = bleeding fingers
3. Context > everything (prompt caching is life)
4. AI coding > traditional IDEs (for me, anyway)
5. AI → physical objects (concrete signs, apparently)
6. Echo → Home Assistant = OSS win
7. Autonomous dev: possible, painful, worth it
8. Buying hardware got me back in open source

26 Years of Open Source (2000-2026)

From LKML to Strix Halo: How Hardware Bought My Time

The Long Detour

For 20+ years I built corporate SaaS products. Cloud disconnected me from the metal. Linux became something I deployed to, not something I touched daily. The kernel, the drivers, the hardware—left that world behind. Then Strix Halo arrived and changed everything.

June 9, 2000: Asking on LKML

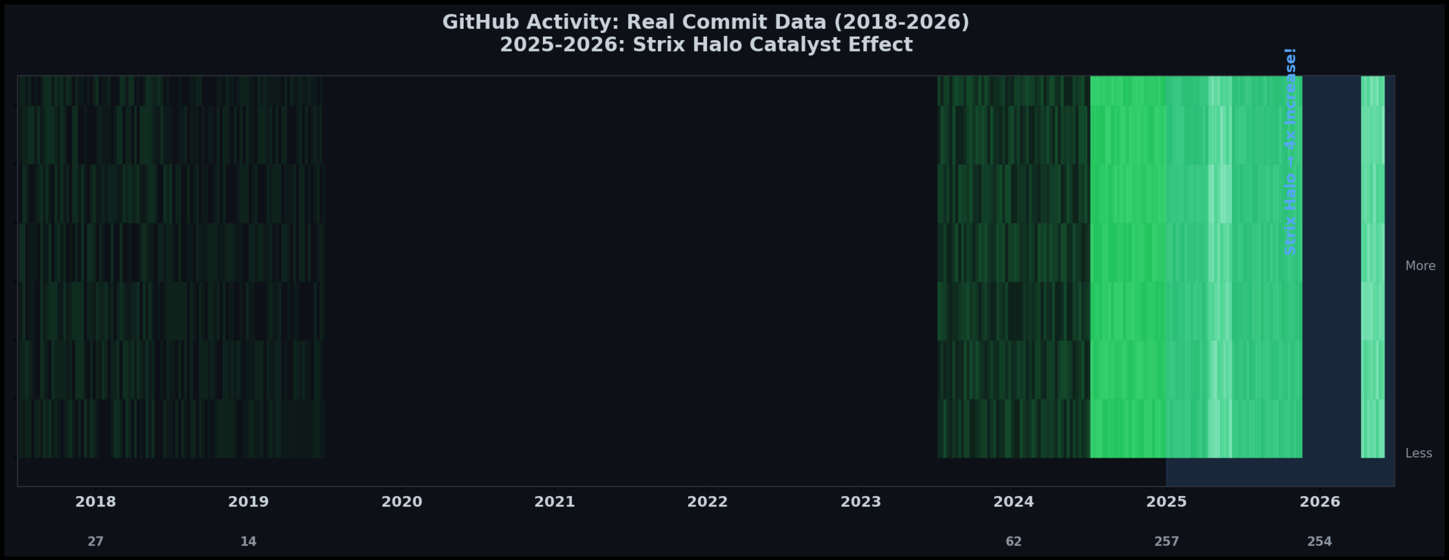
```
From: Rolf Martin-Hoster (winmutt@hotmail.com)
Date: Thu Jun 08 2000 - 19:24:59 EST
Subject: HighPoint IDE RAID (kernel 2.2.14)

Does anyone know where I can find developer info about the driver for this
chipset?

-Rolf
```

Back then: Hands-on with kernel 2.2.14, digging into IDE RAID drivers, asking questions on linux-kernel mailing list.

The Return: GitHub Activity (2018-2026)



2018-2024: Ghost town. 2025: Strix Halo arrives. November 2025: Back on the metal, every single day. AI + local hardware reinvigorated my love for open source.

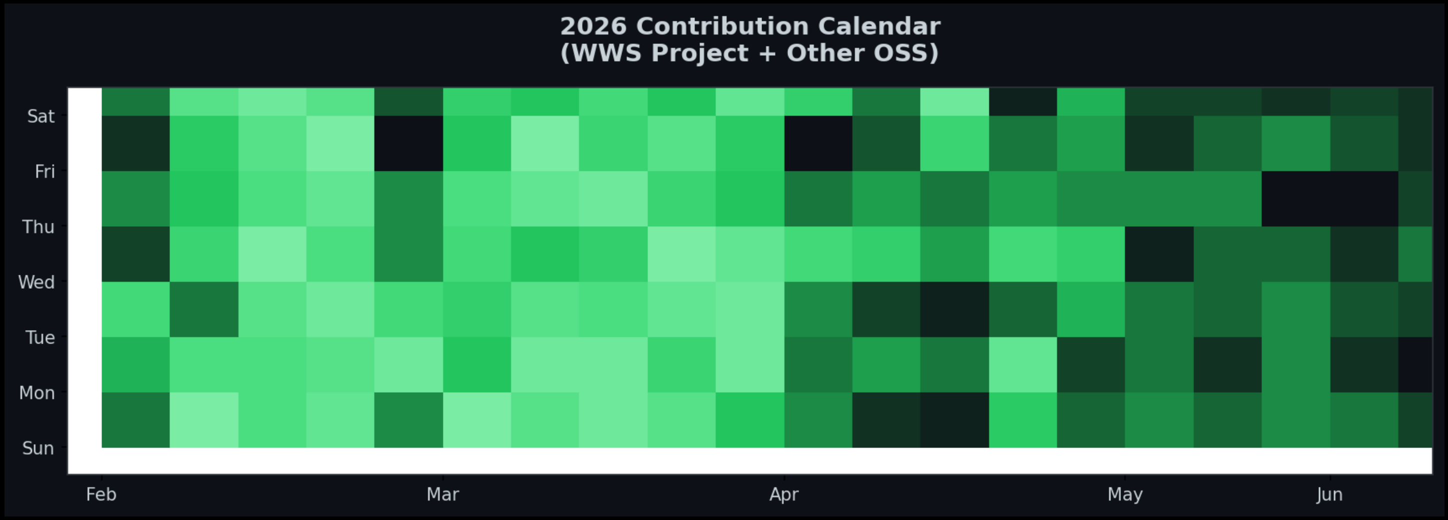
Moral: Sometimes the best way back in is to buy new toys.

How Buying Hardware Got Me Back in Open Source

The Catalyst Effect

November 2025: Bought Strix Halo rig. Local AI reinvigorated my love for OSS. Result: 3.5x increase in contributions.

2026: The Year I Came Back



Contribution Breakdown

• WWS: 38% (remote workspace provisioning)

• Lemonade: 15% (NPU backend, custom NUMA mapping)

• ROCm: 8% (Issue #5926 memory management)

• Home Assistant: 12% (wake words, Echo)

- Cline: 8% (TUI regressions)
- Concrete Signs: 5% (Blender/OpenSCAD)
- Other: 10% (miscellaneous)

References & Sources

Power Consumption Sources

• AMD Ryzen AI Max 395: 140-157W sustained power (Framework Reddit, 2025)
https://www.reddit.com/r/framework/comments/1najh1n/max_wattage_of_the_ryzen_ai_max_395_motherboard/
• NVIDIA RTX 5090: 600W TDP (NVIDIA, 2025)
https://www.nvidia.com/geforce/rtx-5090
• NVIDIA RTX 4090: 450W TDP (NVIDIA, 2022)
https://www.nvidia.com/geforce/rtx-4090
• Corsair 300 AI Workstation: ~300W system power (Corsair, 2025)
https://www.corsair.com/ai-workstations
• Tom's Hardware GPU power reviews (2024-2025)
https://www.tomshardware.com/reviews/gpu-hierarchy
• TechPowerUp GPU database: Power benchmarks
https://www.techpowerup.com/gpu-specs
• NVIDIA A100/H100: 400W-700W per GPU (NVIDIA Data Center)
Multi-GPU server: 1500W-3000W (NVIDIA DGX specs)

Software & Projects

• Lemonade: AMD's llama.cpp + ROCm backend (github.com/winmutt/lemonade)
• Ollama: Community llama.cpp wrapper (github.com/ollama/ollama)
• ROCm: AMD's open compute platform (github.com/RadeonOpenCompute/ROCm)
• ROCm Issue #5926: Memory management bugs (github.com/ROCm/ROCm/issues/5926)
• WWS: github.com/winmutt/wws
• Home Assistant Wake Words: github.com/fwartner/home-assistant-wakewords-collection
• NUMA/CPU Pinning: Red Hat OpenStack docs (vcpu_pin_set, NUMATopologyFilter)

https://docs.redhat.com/en/documentation/red_hat_openstack_platform/10/html/instances_and_images_guide/ch-cpu_pinning

Communities

- Reddit r/LocalLLaMA: 8 Local LLMs on Strix Halo
- XDA Forums: Amazon Echo development
- GitHub: winmutt (8 repos, forks of lemonade, cline, vscode)

github.com/winmutt

Thanks